Inverse potentials of one-body densities, the dual functional approach

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- No spin , static , space $\mathbb{R}^d$, electrons

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- Hamiltonian : operator of $L^2_a\left(\left(\mathbb{R}^d\right)^N, \mathbb{C}\right)$

$$H(v) = \sum_{i=1}^{N} -\Delta_{x_i} + \sum_{1 \leqslant i < j \leqslant N} w(x_i - x_j) + \sum_{i=1}^{N} v(x_i)$$

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- Ground states are given by the eigenspace $\mathrm{Ker}\left(H(v) - E^{(k)}(v)\right)$, where

$$E^{(k)}(v) := \sup_{\substack{A \subset L^2_a(\Omega^N) \\ \dim_{\mathbb{C}} A = k}} \inf_{\substack{\Psi \in A^\perp \\ \int_{\Omega^N} |\Psi|^2 = 1 \\ \Psi \in H^1_a(\Omega^N)}} \langle \Psi, H(v)\Psi \rangle$$

Pure and mixed states

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- Choose a basis $(\Psi_i)_{i\in\mathbb{N}}$. Mixed states are

$$\begin{aligned} \mathrm{Conv} \; & \left\{ P_\Psi, \Psi \in H^1_a(\mathbb{R}^{dN}), \int_{\mathbb{R}^{dN}} |\Psi|^2 = 1 \right\} \\ & = \left\{ \sum_{i\in\mathbb{N}} \lambda_i P_{\Psi_i} \;\Big|\; \sum_{i=1}^{+\infty} \lambda_i = 1, \lambda_i \geqslant 0 \right\} \\ & = \left\{ \Gamma \text{ op of } H^1_a(\mathbb{R}^{dN}) \;\big|\; \Gamma = \Gamma^* \geqslant 0, \mathrm{Tr}\, \Gamma = 1 \right\} \end{aligned}$$

$$\text{ground mixed states : } \operatorname{ran}\Gamma \subset \operatorname{Ker}\left(H(v) - E^{(k)}(v)\right)$$

The one-body density

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$$\rho_\Psi(x) := N \int_{\mathbb{R}^{d(N-1)}} |\Psi|^2 (x, x_2, \ldots, x_N) \mathrm{d}x_2 \cdots \mathrm{d}x_N$$

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- $\rho \geqslant 0$, $\int \rho_\Psi = N$, $\sqrt{\rho} \in H^1$

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- $\Gamma = \Gamma^* \geqslant 0$ with $\operatorname{ran}\Gamma \in \operatorname{Ker}\left(H(v) - E^{(k)}(v)\right)$ such that

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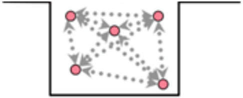

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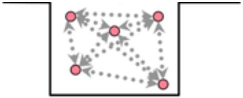

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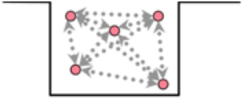

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$$\downarrow$$

$$\Psi^{(k)}, E^{(k)}, \ldots$$

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- Numerical studies (Gavini, ...)

Article

"Building Kohn–Sham potentials for ground and excited states"
(G. 22)

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The dual functional

$$G_\rho^{(k)}(v) := E^{(k)}(v) - \int_{\mathbb{R}^d} v\rho$$

Dual optimality

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- On degenerate potentials, $v \mapsto \rho_{\Psi(v)}$ and $E^{(k)}$ are not differentiable
- Duality with the Levy-Lieb functional

$$F_{\text{mix}}^{(k)}(\rho) := \sup_{\substack{A \subset H^1_a(\Omega^N) \\ \dim_{\mathbb{C}} A = k}} \inf_{\substack{\Gamma \in \mathcal{S}^N_{\text{mix}}(A^\perp, \Omega) \\ \rho_\Gamma = \rho}} \operatorname{Tr} H(0)\Gamma.$$

one has

$$\sup_{v \in (L^p + L^\infty)(\Omega, \mathbb{R})} G_\rho^{(k)}(v) = F_{\text{mix}}^{(k)}(\rho)$$

Dual optimality

Theorem (Optimality in the dual problem)

Take $\rho \in L^1(\mathbb{R}^d)$, $\rho \geqslant 0$, $\int \rho = N$, $\sqrt{\rho} \in H^1(\mathbb{R}^d)$, $v \in L^{p>2}$ such that $H(v)$ has a ground state. Those statements are equivalent

- *there is a k^{th} bound mixed state Γ of v such that $\rho_\Gamma = \rho$*
- *v is a local maximizer of $G_\rho^{(k)}$*
- *v is a global maximizer of $G_\rho^{(k)}$*

Local dual problem

We define $m_k^v, M_k^v \in \mathbb{N}$ by

$$E^{(m_k^v-1)}(v) < E^{(m_k^v)}(v) = \cdots = E^{(k)}(v) = \cdots = E^{(M_k^v)}(v) < E^{(M_k^v+1)}(v),$$

$$^{+}\delta_v G_\rho^{(k)}(u) = \max_{\substack{Q \subset \mathrm{Ker}\left(H(v) - E^{(k)}(v)\right) \\ \dim Q = M_k^v - k + 1}} \min_{\substack{\Psi \in Q \\ \int_{\Omega^N} |\Psi|^2 = 1}} \int_\Omega (\rho_\Psi - \rho)\, u$$

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Proposition (Local dual problem)

$$\sup_{\substack{u\in L^p\\ \|u\|_{L^p}=1}} {}^+\delta_v G_\rho^{(k)}(u) = \max_{\substack{Q\subset \mathrm{Ker}(H(v)-E^{(k)}(v))\\ \dim Q = M_k - k+1}} \min_{\substack{\Gamma\in\mathcal{S}(Q)\\ \Gamma\geqslant 0, \mathrm{Tr}\,\Gamma=1}} \|\rho_\Gamma - \rho\|_{L^{p'}},$$

and the supremum is attained by $u^* = \left|\frac{\rho_{\Gamma^*}-\rho}{\|\rho_{\Gamma^*}-\rho\|_{L^{p'}}}\right|^{p'-1} \mathrm{sgn}(\rho_{\Gamma^*}-\rho)$, *where* Γ^* *is an optimizer of the right hand side.*

Proof

- <u>v is local maximizer $\implies$ v-representability</u>

For u small, we have $G_\rho^{(k)}(v+u) \leqslant G_\rho^{(k)}(v)$ and

$$\sup_{\substack{u \in L^p(\Omega,\mathbb{R}) \\ \|u\|_{L^p}=1}} {}^+\delta_v G_\rho^{(k)}(u) \leqslant 0$$

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- v-representability $\implies$ v is global maximizer

Consider Γ, $\operatorname{ran}\Gamma \subset \operatorname{Ker}\left(H(v) - E^{(k)}(v)\right)$ and $\rho_\Gamma = \rho$, then

$$\mathcal{E}_v(\Gamma) = E^{(k)}(v) = \sup_{\substack{A\subset H^1_{\mathrm{a}}(\Omega^N) \\ \dim A=k+1}} \min_{\Lambda\in\mathcal{S}^N_{\mathrm{mix}}(A^\perp,\Omega)} \mathcal{E}_v(\Lambda)$$

$$= \sup_{\substack{A\subset H^1_{\mathrm{a}}(\Omega^N) \\ \dim A=k+1}} \min_{\substack{\Lambda\in\mathcal{S}^N_{\mathrm{mix}}(A^\perp,\Omega) \\ \rho_\Lambda=\rho}} \mathcal{E}_v(\Lambda) = F^{(k)}_{\mathrm{mix}}(\rho) + \int_\Omega v\rho$$

hence $$G_\rho^{(k)}(v) = F^{(k)}_{\mathrm{mix}}(\rho) = \sup_{u\in L^p(\Omega,\mathbb{R})} G_\rho^{(k)}(u)$$

Dual optimality

Proposition (Pure state v-representability)

Take $\rho \in L^1(\mathbb{R}^d)$, $\rho \geqslant 0$, $\int \rho = N$, $\sqrt{\rho} \in H^1(\mathbb{R}^d)$, $v \in L^{p>2}$ such that $H(v)$ has a ground state. If v maximizes $G_\rho^{(k)}$ and

- $\dim \operatorname{Ker} \left(H(v) - E^{(k)}(v) \right) \in \{1, 2\}$,
- *or $d = 1$ and $w = 0$,*

then v has a k^{th} bound pure state Ψ such that $\rho_\Psi = \rho$.

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Proof.

$$\left\{\rho_\psi \;\middle|\; \psi \in \mathrm{Span}_{\mathbb{C}}\left(\varphi, \phi\right), \int_{\Omega^N} |\psi|^2 = 1\right\}$$
$$= \left\{\rho_\Gamma \;\middle|\; \Gamma \in \mathcal{S}\left(\mathrm{Span}_{\mathbb{R}}(\varphi, \phi)\right), \Gamma \geqslant 0, \mathrm{Tr}\,\Gamma = 1\right\}.$$

$\square$

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Not coercive

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Take $(v_n)_{n\in\mathbb{N}} \in Y^{\mathbb{N}}$ a maximizing sequence, then $\|v_n\|_Y$ is bounded so there is $v_* \in Y$ such that $v_{k_n} \rightharpoonup v_*$. By weak upper semi-continuity of $E^{(k)}$ in Y, v_* maximizes $G_\rho^{(k)}$.

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Counter-examples : $v \in L^1 \cap L^{p>1}$, $v_n(x) := n^\alpha v(nx)$,
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  - if $v \geqslant 0$, then $E^{(k)}(v_n) = 0$ so $G_\rho^{(k)}(v) = -\int v_n\rho \to -\rho(0)\int v$ is bounded

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  - if $v \geqslant 0$, then $E^{(k)}(v_n) = 0$ so $G_\rho^{(k)}(v) = -\int v_n\rho \to -\rho(0)\int v$ is bounded
  - if $v \leqslant 0$, $G_\rho^{(k)}(v_n) \geqslant -L_{1,d} n^{\alpha\left(1+\frac{d}{2}\right)-d} \int_{\mathbb{R}^d} |v|^{1+\frac{d}{2}}$ and we choose $\alpha = d/\left(1+\frac{d}{2}\right)$

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Regularization

- Dual : restriction to potentials $V = \sum_{i\in I} v_i\alpha_i$, $v \in (v_i)_{i\in I} \in \ell^\infty(I,\mathbb{R})$, $\alpha_i \in L^\infty(\Omega)$, $\sum_{i\in I}\alpha_i = \mathbb{1}_\Omega$, $r_i \in \mathbb{R}_+$, $r_i = \int \rho\alpha_i$, $\sum_{i\in I} r_i = N$

$$G^{(k)}_{r,\boldsymbol{\alpha}}(v) := E^{(k)}\left(\sum_{i\in I} v_i\alpha_i\right) - \sum_{i\in I} v_i r_i,$$

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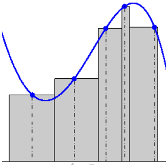

Proof of coercivity

For all test function $\Psi \in H^1$,

$$G_\rho^{(0)}(v) \leqslant \mathcal{E}_0(\Psi) + \int_{\mathbb{R}^d} v(\rho_\Psi - \rho)$$
$$\leqslant \mathcal{E}_0(\Psi) - \left(\int_{\mathbb{R}^d} |v_+| (\rho - \rho_\Psi) + \int_{\mathbb{R}^d} |v_-| (\rho_\Psi - \rho) \right)$$

(Inspired from Chayes, Chayes, Ruskai - 1985)

$$\Psi_{j,Q}(X) := \frac{1}{\sqrt{\sum_{v_i < c_\Omega} \int_\Omega \alpha_i \rho}} \sum_{v_i < c_\Omega} \sqrt{\int_\Omega \alpha_i \rho} \;\; \Phi_j(X - Y_i).$$

$$G_{r,\alpha}^{(k)}(v) \leqslant -\min\left(1, \frac{\sum_{v_i \geqslant c_\Omega} \int_\Omega \alpha_i \rho}{\sum_{v_i < c_\Omega} \int_\Omega \alpha_i \rho}\right) \|v - c_\Omega\|_{\ell^1_r} + c_R,$$

Coercivity

Corollary (Existence of the inverse potential)

When I is finite $G^{(k)}_{r,\alpha}$ is coercive and there exists a maximizer v. If $\Omega \subset \mathbb{R}^d$ is bounded, there is a k^{th} excited N-particle ground mixed state Γ_v of $H\left(\sum_{i\in I} v_i\alpha_i\right)$ such that $\int \alpha_i \rho_{\Gamma_v} = r_i$ $(= \int \alpha_i \rho)$ $\forall i$.

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$$\int \alpha_i \rho_{\Gamma_v} \underset{\forall i}{=} \int \alpha_i \rho, \qquad \Longrightarrow \qquad \rho_{\Gamma_v} \simeq \rho$$

Coercivity

Corollary (Existence of the inverse potential)

When I is finite $G_{r,\alpha}^{(k)}$ is coercive and there exists a maximizer v. If $\Omega \subset \mathbb{R}^d$ is bounded, there is a k^{th} excited N-particle ground mixed state Γ_v of $H\left(\sum_{i \in I} v_i \alpha_i\right)$ such that $\int \alpha_i \rho_{\Gamma_v} = r_i$ $(= \int \alpha_i \rho)$ $\forall i$.

$$\int \alpha_i \rho_{\Gamma_v} \underset{\forall i}{=} \int \alpha_i \rho, \qquad \Longrightarrow \qquad \rho_{\Gamma_v} \simeq \rho$$

$\boxed{\textbf{Constructive inversion with mixed states}}$

For given ρ, $\epsilon > 0$, $k \in \mathbb{N}$ there exists a potential v and Γ_v with $\operatorname{Ran} \Gamma_v \subset \operatorname{Ker}\left(H(v) - E^{(k)}(v)\right)$ such that $\|\rho_{\Gamma_v} - \rho\|_{L^1 \cap L^q} \leqslant \epsilon$.

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"Gradient" ascent

- Minimize $J(v) := \int_{\mathbb{R}^d} \left(\rho_{\Psi(v)} - \rho\right)^2$?

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- Second idea, maximize

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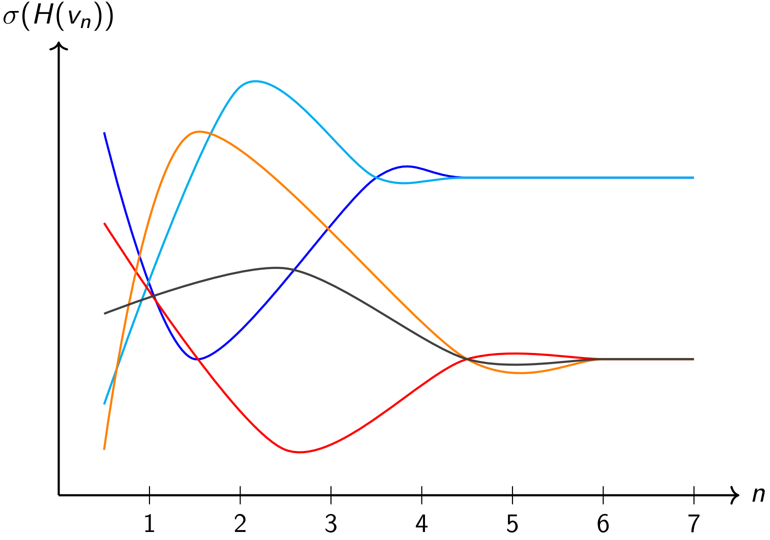
- Line search for α, temperature

Level crossing and blocking



Goal

What we know

- Approximate inversion with mixed states for any k

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- Inversion with pure states for $d = 2$?

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$d = 1$

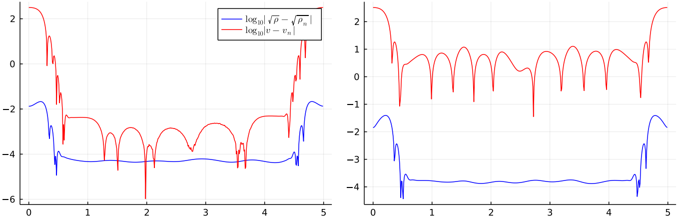

Figure: Plot for $d = 1$, $N = 5$, $k = 0$ on the left, $k = 3$ on the right, $\log_{10}|\rho_n - \rho|$, $\log_{10}|v_n - v|$

Uniqueness

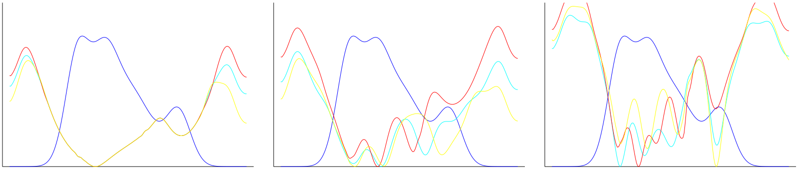

Figure: $d = 1$, $N = 3$, $k = 0$ left, $k = 1$ middle, $k = 5$ right. Densities in blue, inverse potentials in other colors

$d = 3$

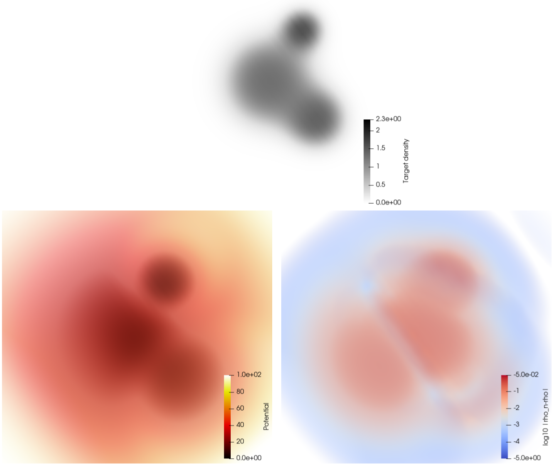

Figure: $d = 3,\ N = 4,\ k = 1\ ;\ \rho,\ v_n,\ \log_{10}|\rho_n - \rho|$

Simulations at high densities

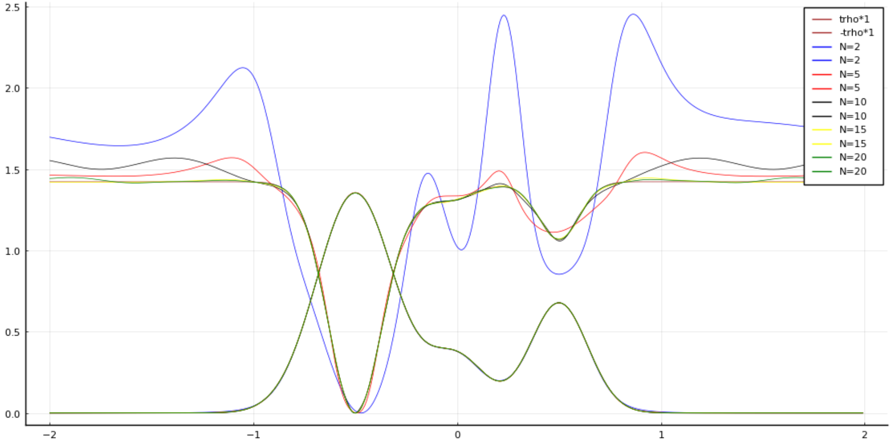

Figure: Convergence of $\rho_N^{-1}(N\rho)/N^{\frac{2}{d}}$, $\int \rho = 1$

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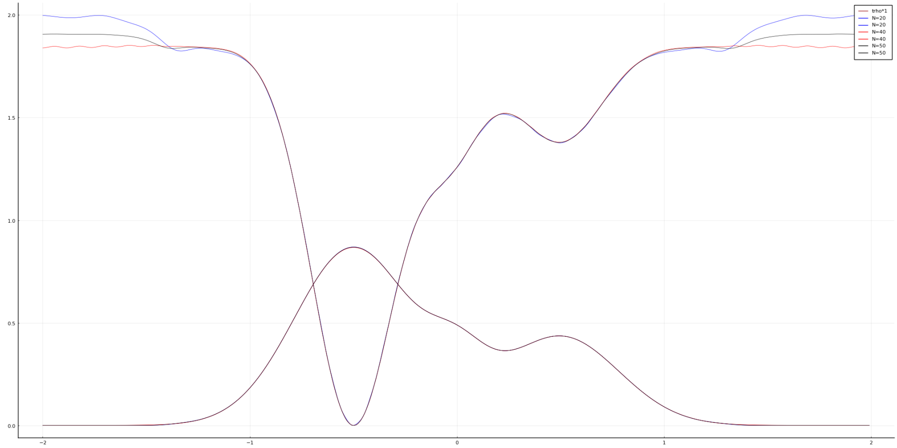

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Conjecture

For any $\rho \geqslant 0$ such that $\int \rho = 1$ and $\sqrt{\rho} \in H^1$,

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- Degeneracies are generic, even for $d = 1$. Need to be considered, not in literature

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- Treatment of degeneracies in the algorithm